FROM NONISOTHERMAL BGK TO EULER MAXWELLIANS VIA RELATIVE ENTROPY

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Abstract. We study the hydrodynamic limit of the nonisothermal BGK model toward smooth Euler Maxwellians. For a prescribed smooth Euler solution, we derive a relative entropy stability estimate between a BGK solution and the associated Maxwellian. The main new ingredient is the control of an additional velocity-cubic term in the relative entropy identity. Under a uniform sixth velocity-moment bound and suitable bounds on the BGK macroscopic quantities, we obtain a uniform-in-time relative entropy estimate. For well-prepared initial data, this yields strong L^1 convergence of the BGK solution and the local Maxwellians to the target Euler Maxwellian, together with convergence of the associated macroscopic quantities.

1. Introduction

The Bhatnagar–Gross–Krook (BGK) model, introduced in [6], is a kinetic relaxation equation in which the distribution function $f^\varepsilon(t,x,v)$ relaxes, on the time scale $\varepsilon > 0$, towards a local Maxwellian determined by its conserved moments. In this paper we consider the nonisothermal BGK equation on $[0,T) \times \mathbb{R}^d_x \times \mathbb{R}^d_v$,

$$\partial_t f^\varepsilon + v \cdot \nabla_x f^\varepsilon = -\frac{1}{\varepsilon}\big(f^\varepsilon - \mathcal{M}(f^\varepsilon)\big), \tag{1.1}$$

where $\mathcal{M}(f^\varepsilon)$ is the local Maxwellian with the same mass, momentum, and kinetic energy as f^ε:

$$\mathcal{M}(f^\varepsilon)(t,x,v) = \frac{1}{(2\pi\theta^\varepsilon(t,x))^{d/2}}\rho^\varepsilon(t,x)\exp\left(-\frac{|v-u^\varepsilon(t,x)|^2}{2\theta^\varepsilon(t,x)}\right). \tag{1.2}$$

The corresponding density, mean velocity, and temperature are given by

$$\rho^\varepsilon(t,x) = \int_{\mathbb{R}^d} f^\varepsilon(t,x,v)\,\mathrm{d}v, \tag{1.3}$$

$$u^\varepsilon(t,x) = \begin{cases} \dfrac{1}{\rho^\varepsilon(t,x)}\displaystyle\int_{\mathbb{R}^d} v\, f^\varepsilon(t,x,v)\,\mathrm{d}v, & \text{if } \rho^\varepsilon(t,x) > 0,\\ 0, & \text{if } \rho^\varepsilon(t,x) = 0, \end{cases} \tag{1.4}$$

and

$$\theta^\varepsilon(t,x) = \begin{cases} \dfrac{1}{d\,\rho^\varepsilon(t,x)}\displaystyle\int_{\mathbb{R}^d} |v-u^\varepsilon(t,x)|^2 f^\varepsilon(t,x,v)\,\mathrm{d}v, & \text{if } \rho^\varepsilon(t,x) > 0,\\ 1, & \text{if } \rho^\varepsilon(t,x) = 0. \end{cases} \tag{1.5}$$

By construction, $\mathcal{M}(f^\varepsilon)$ shares with f^ε the collision invariants 1, v, and $|v|^2$, so the relaxation term preserves mass, momentum, and kinetic energy.

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Formally, as $\varepsilon \to 0$, one expects f^ε to approach local Maxwellians and the corresponding moments to satisfy the compressible Euler system with pressure law $p = \rho\theta$,

$$\partial_t \rho + \nabla_x \cdot (\rho u) = 0, \qquad \partial_t(\rho u) + \nabla_x \cdot (\rho u \otimes u) + \nabla_x(\rho\theta) = 0, \tag{1.6}$$

$$\partial_t E + \nabla_x \cdot \big(u(E + \rho\theta)\big) = 0, \qquad E = \frac{1}{2}\rho|u|^2 + \frac{d}{2}\rho\theta, \tag{1.7}$$

at least before shock formation, that is, on time intervals where (ρ, u, θ) remains smooth.

The purpose of this paper is to justify this hydrodynamic limit in the smooth regime by the relative entropy method. Given a smooth Euler solution (ρ, u, θ) of (1.6)–(1.7) on $[0, T) \times \mathbb{R}^d$ with strictly positive density and temperature, we consider the associated Maxwellian

$$M(t, x, v) = \frac{1}{(2\pi\theta(t, x))^{d/2}} \rho(t, x) \exp\left(-\frac{|v - u(t, x)|^2}{2\theta(t, x)}\right). \tag{1.8}$$

For nonnegative phase-space densities f, g of unit mass and with $g > 0$, we use the relative entropy

$$H(f \mid g) := \iint_{\mathbb{R}^{2d}} f \log \frac{f}{g} \, \mathrm{d}v \, \mathrm{d}x, \tag{1.9}$$

with the convention $0 \log 0 := 0$.

A central point in the argument is the computation of $(\partial_t + v \cdot \nabla_x) \log M$. The resulting relative entropy identity contains a traceless stress term and an additional velocity-cubic contribution. Our argument shows that, under a sixth velocity-moment bound and suitable uniform bounds on the BGK macroscopic quantities, this term can still be controlled in the smooth regime. We derive a Grönwall-type estimate for $H(f^\varepsilon \mid M)$ and deduce strong convergence, uniform in time, of f^ε towards M in $L^1(\mathbb{R}^{2d})$, together with strong convergence of the local Maxwellians $\mathcal{M}(f^\varepsilon)$ towards M. We also obtain the convergence of the associated macroscopic quantities. The precise assumptions and the statement of the main theorem are given in Section 2.

The use of relative entropy in stability theory goes back to Dafermos [9] and was developed systematically for fluid-dynamic limits and hyperbolic relaxation by Tzavaras; see, for instance, [14, 15]. In the kinetic setting, Berthelin and Vasseur [4] established, by a relative entropy argument, the convergence from BGK models to multidimensional Euler equations before shock formation for pressure laws of the form $p(\rho) = \rho^\gamma$ with $\gamma > 1$. The present work follows the same general strategy in the nonisothermal setting, where the pressure is given by $p(\rho, \theta) = \rho\theta$ and the temperature evolves as an additional unknown.

Related works by Berthelin and Bouchut [1, 2, 3] study BGK systems relaxing to gas-dynamic models by means of finite-energy, kinetic-entropy, and invariant-domain methods. A discrete-velocity counterpart was investigated by Berthelin, Tzavaras, and Vasseur [5]. More recently, Choi and Hwang [7], Hwang [11], and Koo and Song [12] established weak, classical, and mild solution theories for related BGK-type models. See also Choi and Hwang [8] for a BGK-type hydrodynamic limit with alignment interactions. For existence and stability results for the BGK equation itself, we also refer to Perthame [13].

The present paper adapts the relative entropy method to the nonisothermal BGK model and identifies the additional heat-flux term that appears in the associated relative entropy identity. To the best of our knowledge, the rigorous BGK hydrodynamic-limit results closest to the present work concern regimes in which the pressure depends only on the density, whereas the full nonisothermal BGK limit to Euler Maxwellians has not been treated in this form.

The paper is organized as follows. In Section 2 we collect the assumptions and state the main theorem. Section 3 discusses convergence to Maxwellians. In Section 4 we derive the relative entropy identity. Section 5 gathers a few auxiliary lemmas used in the proof of the main estimate. Finally, in Sections 6 and 7 we prove the theorem and its corollary.

2. Assumptions and main theorem

We now collect the assumptions used in the proof of the hydrodynamic limit and state the main result. Throughout the paper, (ρ, u, θ) denotes a smooth solution of (1.6)–(1.7) on $[0,T) \times \mathbb{R}^d$, and M denotes the associated Maxwellian (1.8). For brevity, we write

$$\|\cdot\|_\infty := \|\cdot\|_{L^\infty((0,T)\times\mathbb{R}^d)}.$$

The argument relies on the following two groups of hypotheses.

(A) *Uniform bounds for the BGK family.* There exist constants $U > 0$, $0 < \underline{\Theta} \le \overline{\Theta} < \infty$, and $A > 0$, independent of ε, such that

$$\|u^\varepsilon\|_\infty \le U,$$

$$0 < \underline{\Theta} \le \theta^\varepsilon(t,x) \le \overline{\Theta} \quad \text{for all } (t,x) \in [0,T) \times \mathbb{R}^d,$$

and

$$\sup_{t\in[0,T)} \iint_{\mathbb{R}^{2d}} |v|^6 f^\varepsilon(t,x,v)\,\mathrm{d}v\,\mathrm{d}x \le A.$$

(B) *Smooth Euler background.* The Euler solution (ρ, u, θ) has unit mass,

$$\int_{\mathbb{R}^d} \rho(t,x)\,\mathrm{d}x = 1 \qquad \text{for all } t \in [0,T),$$

strictly positive density, and temperature bounded away from zero and infinity:

$$\rho(t,x) > 0, \qquad 0 < \underline{\theta} \le \theta(t,x) \le \overline{\theta} \qquad \text{for all } (t,x) \in [0,T) \times \mathbb{R}^d.$$

Moreover,

$$u,\ \nabla_x u,\ \nabla_x \log\theta \in L^\infty((0,T)\times\mathbb{R}^d).$$

Throughout the paper, we further assume that the smooth BGK solutions under consideration decay sufficiently fast at infinity in (x,v) so that all integrations by parts and differentiations under the integral sign are justified.

Theorem 2.1. *Let f^ε be a smooth solution of the BGK equation* (1.1) *on* $[0,T) \times \mathbb{R}^d_x \times \mathbb{R}^d_v$ *with $f^\varepsilon \ge 0$ and unit total mass*

$$\iint_{\mathbb{R}^{2d}} f^\varepsilon(t,x,v)\,\mathrm{d}v\,\mathrm{d}x = 1 \qquad \textit{for all } t \in [0,T).$$

Assume that (A) *and* (B) *hold. Then there exists a positive constant*

$$C = C\big(d, \underline{\theta}, \overline{\theta}, U, \underline{\Theta}, \overline{\Theta}, A, \|u\|_\infty, \|\nabla_x u\|_\infty, \|\nabla_x \log\theta\|_\infty\big),$$

independent of ε*, such that*

$$\frac{\mathrm{d}}{\mathrm{d}t} H(f^\varepsilon \mid M) \le C\big(H(f^\varepsilon \mid M) + \varepsilon\big) \qquad \text{for all } t \in [0,T). \tag{2.1}$$

Consequently, if at $t = 0$*,*

$$H(f_0^\varepsilon \mid M_0) \to 0, \qquad \text{as } \varepsilon \to 0, \tag{2.2}$$

then

$$\sup_{t\in[0,T)} H\big(f^\varepsilon(t) \mid M(t)\big) \to 0, \qquad \text{as } \varepsilon \to 0. \tag{2.3}$$

Theorem 2.1 provides the relative entropy stability estimate and, under well-prepared initial data, yields uniform-in-time convergence in relative entropy. The next corollary records the corresponding kinetic and macroscopic consequences.

Corollary 2.2. *Under the hypotheses of Theorem 2.1, together with* (2.2)*, one has*

$$f^\varepsilon \to M, \qquad \mathcal{M}(f^\varepsilon) \to M \qquad \text{in } L^\infty(0,T; L^1(\mathbb{R}^{2d})), \tag{2.4}$$

as well as

$$\rho^\varepsilon \to \rho, \qquad \rho^\varepsilon u^\varepsilon \to \rho u, \qquad \rho^\varepsilon \theta^\varepsilon \to \rho\theta \qquad \text{in } L^\infty(0,T; L^1(\mathbb{R}^d)). \tag{2.5}$$

3. Convergence to Maxwellians

In this section we collect a few relative entropy identities involving Maxwellians that will be used repeatedly in the proof of the main theorem. In particular, they provide the link between convergence in relative entropy, strong convergence in phase space, and convergence of the associated macroscopic quantities.

We consider the space

$$\mathcal{F} = \Big\{ f \in L^1(\mathbb{R}^{2d}) : \ f \ge 0, \ \|f\|_{L^1(\mathbb{R}^{2d})} = 1, \ \iint_{\mathbb{R}^{2d}} |v|^2 f(x,v)\,\mathrm{d}x\,\mathrm{d}v < \infty \Big\},$$

and for each $f \in \mathcal{F}$ we define its density, mean velocity and temperature respectively by

$$\rho_f(x) := \int_{\mathbb{R}^d} f(x,v)\,\mathrm{d}v,$$

$$u_f(x) := \begin{cases} \dfrac{1}{\rho_f(x)} \displaystyle\int_{\mathbb{R}^d} v f(x,v)\,\mathrm{d}v, & \text{if } \rho_f(x) > 0, \\ 0, & \text{if } \rho_f(x) = 0, \end{cases}$$

and

$$\theta_f(x) := \begin{cases} \dfrac{1}{d\,\rho_f(x)} \displaystyle\int_{\mathbb{R}^d} |v - u_f(x)|^2 f(x,v)\,\mathrm{d}v, & \text{if } \rho_f(x) > 0, \\ 1, & \text{if } \rho_f(x) = 0. \end{cases}$$

The associated local Maxwellian is

$$\mathcal{M}(f)(x,v) = \frac{\rho_f(x)}{(2\pi\theta_f(x))^{d/2}} \exp\left(-\frac{|v - u_f(x)|^2}{2\theta_f(x)} \right).$$

We say that f is a Maxwellian if $f = \mathcal{M}(f)$.

For $f, g \in \mathcal{F}$, the Csiszár–Kullback–Pinsker inequality (see [10]) states that

$$\|f - g\|^2_{L^1(\mathbb{R}^{2d})} \le 2\, H(f \mid g). \tag{3.1}$$

The first lemma gives the basic entropy splitting associated with the Maxwellian projection.

Lemma 3.1. *For $f, g \in \mathcal{F}$, whenever the entropies are finite,*

$$H(g \mid \mathcal{M}(f)) = H(g \mid \mathcal{M}(g)) + H(\mathcal{M}(g) \mid \mathcal{M}(f)). \tag{3.2}$$

Proof. We write

$$H(g \mid \mathcal{M}(f)) = \iint_{\mathbb{R}^{2d}} g \log \frac{g}{\mathcal{M}(g)} \, \mathrm{d}v \, \mathrm{d}x + \iint_{\mathbb{R}^{2d}} g \log \frac{\mathcal{M}(g)}{\mathcal{M}(f)} \, \mathrm{d}v \, \mathrm{d}x.$$

Since $\mathcal{M}(g)$ and $\mathcal{M}(f)$ are Maxwellians, a direct computation shows that

$$\log \frac{\mathcal{M}(g)}{\mathcal{M}(f)} = a(x) + b(x) \cdot v + c(x)|v|^2$$

for suitable scalar functions a, c and vector field b. Because g and $\mathcal{M}(g)$ have the same moments against the collision invariants 1, v, and $|v|^2$, we obtain

$$\begin{aligned} \iint_{\mathbb{R}^{2d}} g \log \frac{\mathcal{M}(g)}{\mathcal{M}(f)} \, \mathrm{d}v \, \mathrm{d}x &= \iint_{\mathbb{R}^{2d}} \mathcal{M}(g) \log \frac{\mathcal{M}(g)}{\mathcal{M}(f)} \, \mathrm{d}v \, \mathrm{d}x \\ &= H(\mathcal{M}(g) \mid \mathcal{M}(f)). \end{aligned}$$

This proves (3.2). □

We next record the explicit formula for the relative entropy between two Maxwellians. This identity will later allow us to extract quantitative information on the density, velocity, and temperature errors.

Lemma 3.2. *Let $M_1 = M_{\rho_1, u_1, \theta_1}$ and $M_2 = M_{\rho_2, u_2, \theta_2}$ be Maxwellians of the form* (1.8)*, with*

$$\rho_1 \geq 0, \qquad \rho_2 > 0, \qquad \theta_1 > 0, \qquad \theta_2 > 0.$$

Then

$$\begin{aligned} H(M_1 \mid M_2) = &\int_{\mathbb{R}^d} \rho_1 \log \frac{\rho_1}{\rho_2} \, \mathrm{d}x + \frac{1}{2} \int_{\mathbb{R}^d} \rho_1 \frac{|u_1 - u_2|^2}{\theta_2} \, \mathrm{d}x \\ &+ \frac{d}{2} \int_{\mathbb{R}^d} \rho_1 \left(\frac{\theta_1}{\theta_2} - 1 - \log \frac{\theta_1}{\theta_2} \right) \mathrm{d}x, \end{aligned} \tag{3.3}$$

with the convention $0 \log 0 := 0$.

Proof. We first note that on $\{\rho_1 = 0\}$ we have $M_1(x, v) = 0$ for every $v \in \mathbb{R}^d$, and therefore both sides of (3.3) vanish there. It is thus enough to work on $\{\rho_1 > 0\}$.

A direct computation gives

$$\log \frac{M_1}{M_2} = \log \frac{\rho_1}{\rho_2} - \frac{d}{2} \log \frac{\theta_1}{\theta_2} - \frac{|v - u_1|^2}{2\theta_1} + \frac{|v - u_2|^2}{2\theta_2}.$$

Integrating against M_1 and using

$$\int_{\mathbb{R}^d} |v - u_1|^2 M_1 \, \mathrm{d}v = d\rho_1\theta_1, \qquad \int_{\mathbb{R}^d} |v - u_2|^2 M_1 \, \mathrm{d}v = d\rho_1\theta_1 + \rho_1|u_1 - u_2|^2,$$

we obtain (3.3). □

As a first consequence of the entropy splitting, convergence in relative entropy towards a Maxwellian implies strong convergence of the associated Maxwellians.

Proposition 3.3. *Let $\{f_n\}_{n\in\mathbb{N}} \subseteq \mathcal{F}$ be a sequence and let $f \in \mathcal{F}$ satisfy $f = \mathcal{M}(f)$. If*

$$H(f_n \mid f) \to 0 \qquad \text{as } n \to \infty,$$

then

$$\mathcal{M}(f_n) \to f \qquad \text{in } L^1(\mathbb{R}^{2d}) \qquad \text{as } n \to \infty.$$

Proof. Since $f = \mathcal{M}(f)$, Lemma 3.1 with $g = f_n$ and f gives

$$H(f_n \mid f) = H(f_n \mid \mathcal{M}(f_n)) + H(\mathcal{M}(f_n) \mid f).$$

In particular,

$$H(\mathcal{M}(f_n) \mid f) \le H(f_n \mid f) \to 0 \qquad \text{as } n \to \infty.$$

The conclusion follows from the Csiszár–Kullback–Pinsker inequality (3.1). □

The next proposition gives the converse mechanism needed later in the proof of the main theorem: if the associated Maxwellians converge in relative entropy and the phase-space densities become asymptotically Maxwellian, then the full distributions converge strongly in phase space.

Proposition 3.4. *Let $\{f_n\}_{n\in\mathbb{N}} \subseteq \mathcal{F}$ be a sequence and let $f \in \mathcal{F}$ satisfy $f = \mathcal{M}(f)$. Assume that*

(i) $H(\mathcal{M}(f_n) \mid \mathcal{M}(f)) \to 0$ *as* $n \to \infty$,
(ii) $H(f_n \mid \mathcal{M}(f_n)) \to 0$ *as* $n \to \infty$.

Then

$$f_n \to f \qquad \text{in } L^1(\mathbb{R}^{2d}) \quad \text{as } n \to \infty.$$

Proof. By Lemma 3.1 and the fact that $f = \mathcal{M}(f)$,

$$H(f_n \mid f) = H(f_n \mid \mathcal{M}(f_n)) + H(\mathcal{M}(f_n) \mid \mathcal{M}(f)).$$

The right-hand side converges to zero by assumptions (i)–(ii), hence $H(f_n \mid f) \to 0$. The Csiszár–Kullback–Pinsker inequality (3.1) then implies $f_n \to f$ in $L^1(\mathbb{R}^{2d})$. □

4. Relative entropy identity

In this section we derive the relative entropy identity that lies at the core of the proof of Theorem 2.1. The main step is to compute $(\partial_t + v \cdot \nabla_x) \log M$, where M is the target Maxwellian. Once this computation is available, the relative entropy identity follows by a standard differentiation argument.

Lemma 4.1. *Let (ρ, u, θ) be a smooth solution of* (1.6)–(1.7)*, let M be given by* (1.8)*, and set $c := v - u(t,x)$. Then*

$$\begin{aligned}(\partial_t + v \cdot \nabla_x) \log M = {} & \frac{1}{\theta}\left(c \otimes c - \frac{|c|^2}{d} I_d\right) : \nabla_x u \\ & + \frac{1}{2}\, c \cdot \nabla_x \log\theta \left(\frac{|c|^2}{\theta} - (d+2)\right).\end{aligned} \tag{4.1}$$

Proof. Set $D_u := \partial_t + u \cdot \nabla_x$. The Euler system implies

$$D_u \log\rho = -\nabla_x \cdot u, \qquad D_u u = -\nabla_x \theta - \theta \nabla_x \log\rho,$$

and

$$D_u \log\theta = -\frac{2}{d} \nabla_x \cdot u.$$

Since $\partial_t + v \cdot \nabla_x = D_u + c \cdot \nabla_x$ and

$$\log M = \log \rho - \frac{d}{2} \log(2\pi\theta) - \frac{|c|^2}{2\theta},$$

we compute

$$\begin{aligned}(\partial_t + v \cdot \nabla_x) \log \rho &= -\nabla_x \cdot u + c \cdot \nabla_x \log \rho, \\ -\frac{d}{2}(\partial_t + v \cdot \nabla_x) \log \theta &= \nabla_x \cdot u - \frac{d}{2}\, c \cdot \nabla_x \log \theta.\end{aligned}$$

Moreover,

$$\begin{aligned}(\partial_t + v \cdot \nabla_x) c &= -(\partial_t + v \cdot \nabla_x) u \\ &= \nabla_x \theta + \theta \nabla_x \log \rho - (c \cdot \nabla_x) u,\end{aligned}$$

so that

$$\begin{aligned}-(\partial_t + v \cdot \nabla_x) \frac{|c|^2}{2\theta} &= -\frac{1}{\theta}\, c \cdot \big(\nabla_x \theta + \theta \nabla_x \log \rho - (c \cdot \nabla_x) u\big) \\ &\quad + \frac{|c|^2}{2\theta} (\partial_t + v \cdot \nabla_x) \log \theta \\ &= -c \cdot \nabla_x \log \theta - c \cdot \nabla_x \log \rho + \frac{1}{\theta} c \otimes c : \nabla_x u \\ &\quad - \frac{|c|^2}{d\theta} \nabla_x \cdot u + \frac{|c|^2}{2\theta} c \cdot \nabla_x \log \theta.\end{aligned}$$

Adding the three contributions, the terms involving $c \cdot \nabla_x \log \rho$ cancel and we obtain

$$\begin{aligned}(\partial_t + v \cdot \nabla_x) \log M &= \frac{1}{\theta} c \otimes c : \nabla_x u - \frac{|c|^2}{d\theta} \nabla_x \cdot u \\ &\quad + \frac{1}{2} c \cdot \nabla_x \log \theta \left(\frac{|c|^2}{\theta} - (d+2) \right),\end{aligned}$$

which is exactly (4.1). □

We now insert the formula from Lemma 4.1 into the time derivative of the relative entropy and obtain the key identity for $H(f^\varepsilon \mid M)$.

Proposition 4.2. *Let f^ε be a smooth solution of* (1.1)*, let (ρ, u, θ) be a smooth solution of* (1.6)–(1.7) *with associated Maxwellian M given by* (1.8)*, and set $c := v - u(t,x)$. Then*

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t} H(f^\varepsilon \mid M) &= -\frac{1}{\varepsilon} \iint_{\mathbb{R}^{2d}} \big(f^\varepsilon - \mathcal{M}(f^\varepsilon)\big) \log \frac{f^\varepsilon}{\mathcal{M}(f^\varepsilon)} \,\mathrm{d}v \,\mathrm{d}x \\ &\quad - \int_{\mathbb{R}^d} \frac{1}{\theta} \left(\int_{\mathbb{R}^d} \left(c \otimes c - \frac{|c|^2}{d} I_d \right) f^\varepsilon \,\mathrm{d}v \right) : \nabla_x u \,\mathrm{d}x \\ &\quad - \frac{1}{2} \int_{\mathbb{R}^d} \nabla_x \log \theta \cdot \left(\int_{\mathbb{R}^d} c \left(\frac{|c|^2}{\theta} - (d+2) \right) f^\varepsilon \,\mathrm{d}v \right) \mathrm{d}x.\end{aligned} \tag{4.2}$$

Proof. We differentiate $H(f^\varepsilon \mid M)$ in time and obtain

$$\frac{\mathrm{d}}{\mathrm{d}t} H(f^\varepsilon \mid M) = \iint_{\mathbb{R}^{2d}} \partial_t f^\varepsilon \, \log \frac{f^\varepsilon}{M} \,\mathrm{d}v \,\mathrm{d}x + \iint_{\mathbb{R}^{2d}} f^\varepsilon \, \partial_t \log \frac{f^\varepsilon}{M} \,\mathrm{d}v \,\mathrm{d}x. \tag{4.3}$$

Using the BGK equation in the first term, we find

$$\begin{aligned}\iint_{\mathbb{R}^{2d}} \partial_t f^\varepsilon \log \frac{f^\varepsilon}{M}\,\mathrm{d}v\,\mathrm{d}x = &-\frac{1}{\varepsilon}\iint_{\mathbb{R}^{2d}} \big(f^\varepsilon - \mathcal{M}(f^\varepsilon)\big)\log\frac{f^\varepsilon}{M}\,\mathrm{d}v\,\mathrm{d}x \\ &- \iint_{\mathbb{R}^{2d}} (v\cdot\nabla_x f^\varepsilon)\log\frac{f^\varepsilon}{M}\,\mathrm{d}v\,\mathrm{d}x.\end{aligned} \tag{4.4}$$

Now

$$\log\frac{f^\varepsilon}{M} = \log\frac{f^\varepsilon}{\mathcal{M}(f^\varepsilon)} + \log\frac{\mathcal{M}(f^\varepsilon)}{M},$$

and, as in the proof of Lemma 3.1,

$$\log\frac{\mathcal{M}(f^\varepsilon)}{M} = a_0(t,x) + b_0(t,x)\cdot v + c_0(t,x)|v|^2$$

for suitable scalar functions a_0, c_0 and vector field b_0. Since f^ε and $\mathcal{M}(f^\varepsilon)$ have the same moments against these collision invariants, we get

$$\iint_{\mathbb{R}^{2d}} \big(f^\varepsilon - \mathcal{M}(f^\varepsilon)\big)\log\frac{\mathcal{M}(f^\varepsilon)}{M}\,\mathrm{d}v\,\mathrm{d}x = 0.$$

Integrating by parts in x in the transport term gives

$$\begin{aligned}\iint_{\mathbb{R}^{2d}} \partial_t f^\varepsilon \log \frac{f^\varepsilon}{M}\,\mathrm{d}v\,\mathrm{d}x = &-\frac{1}{\varepsilon}\iint_{\mathbb{R}^{2d}} \big(f^\varepsilon - \mathcal{M}(f^\varepsilon)\big)\log\frac{f^\varepsilon}{\mathcal{M}(f^\varepsilon)}\,\mathrm{d}v\,\mathrm{d}x \\ &- \iint_{\mathbb{R}^{2d}} v f^\varepsilon\cdot\nabla_x \log M\,\mathrm{d}v\,\mathrm{d}x.\end{aligned} \tag{4.5}$$

For the second term in (4.3), conservation of total mass yields

$$\begin{aligned}\iint_{\mathbb{R}^{2d}} f^\varepsilon\,\partial_t \log\frac{f^\varepsilon}{M}\,\mathrm{d}v\,\mathrm{d}x &= \iint_{\mathbb{R}^{2d}} f^\varepsilon\,\partial_t(\log f^\varepsilon - \log M)\,\mathrm{d}v\,\mathrm{d}x \\ &= \frac{\mathrm{d}}{\mathrm{d}t}\iint_{\mathbb{R}^{2d}} f^\varepsilon\,\mathrm{d}v\,\mathrm{d}x - \iint_{\mathbb{R}^{2d}} f^\varepsilon\,\partial_t\log M\,\mathrm{d}v\,\mathrm{d}x \\ &= -\iint_{\mathbb{R}^{2d}} f^\varepsilon\,\partial_t\log M\,\mathrm{d}v\,\mathrm{d}x.\end{aligned} \tag{4.6}$$

Combining (4.5) and (4.6) gives

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}H(f^\varepsilon \mid M) = &-\frac{1}{\varepsilon}\iint_{\mathbb{R}^{2d}} \big(f^\varepsilon - \mathcal{M}(f^\varepsilon)\big)\log\frac{f^\varepsilon}{\mathcal{M}(f^\varepsilon)}\,\mathrm{d}v\,\mathrm{d}x \\ &- \iint_{\mathbb{R}^{2d}} f^\varepsilon(\partial_t + v\cdot\nabla_x)\log M\,\mathrm{d}v\,\mathrm{d}x.\end{aligned} \tag{4.7}$$

The identity (4.2) now follows by substituting (4.1) from Lemma 4.1. □

5. Auxiliary lemmas

In this section we record three auxiliary lemmas used in the proof of Theorem 2.1. They concern, respectively, the control of a weighted L^2-distance by the symmetrized relative entropy, the computation of certain Maxwellian moments, and a quadratic estimate for $r - 1 - \log r$ on bounded intervals.

Lemma 5.1. *For $a, b > 0$,*

$$\frac{2(a-b)^2}{a+b} \le (a-b)\log\frac{a}{b}. \tag{5.1}$$

Consequently, for nonnegative f, g with $f + g > 0$ a.e.,

$$\begin{aligned}\iint_{\mathbb{R}^{2d}} \frac{(f-g)^2}{f+g}\,\mathrm{d}v\,\mathrm{d}x &\le \frac{1}{2}\iint_{\mathbb{R}^{2d}} (f-g)\log\frac{f}{g}\,\mathrm{d}v\,\mathrm{d}x \\ &= \frac{1}{2}\big(H(f \mid g) + H(g \mid f)\big).\end{aligned} \tag{5.2}$$

Proof. Since

$$\log\frac{a}{b} = (a-b)\int_0^1 \frac{1}{\tau a + (1-\tau)b}\,\mathrm{d}\tau,$$

we have

$$(a-b)\log\frac{a}{b} = (a-b)^2\int_0^1 \frac{1}{\tau a + (1-\tau)b}\,\mathrm{d}\tau.$$

The function $r \mapsto 1/r$ is convex on $(0,\infty)$, hence Jensen's inequality yields

$$\int_0^1 \frac{1}{\tau a + (1-\tau)b}\,\mathrm{d}\tau \ge \frac{1}{\int_0^1 (\tau a + (1-\tau)b)\,\mathrm{d}\tau} = \frac{2}{a+b}.$$

This proves (5.1). Integrating the pointwise inequality with $a = f$ and $b = g$ gives (5.2). □

Lemma 5.2. *Let $G = \mathcal{M}(f^\varepsilon) = M_{\rho^\varepsilon,u^\varepsilon,\theta^\varepsilon}$ and let $M = M_{\rho,u,\theta}$ be the target Maxwellian. Set $c := v - u(t,x)$ and $\delta u := u^\varepsilon - u$. Then*

$$\int_{\mathbb{R}^d}\left(c\otimes c - \frac{|c|^2}{d}I_d\right)G\,\mathrm{d}v = \rho^\varepsilon\left(\delta u\otimes\delta u - \frac{|\delta u|^2}{d}I_d\right), \tag{5.3}$$

and

$$\int_{\mathbb{R}^d} c\left(\frac{|c|^2}{\theta} - (d+2)\right)G\,\mathrm{d}v = \rho^\varepsilon\,\delta u\left((d+2)\left(\frac{\theta^\varepsilon}{\theta} - 1\right) + \frac{|\delta u|^2}{\theta}\right). \tag{5.4}$$

Proof. Write $c = (v - u^\varepsilon) + (u^\varepsilon - u) =: \omega + \delta u$. Under the Maxwellian G we have

$$\int_{\mathbb{R}^d}\omega\,G\,\mathrm{d}v = 0, \qquad \int_{\mathbb{R}^d}\omega\otimes\omega\,G\,\mathrm{d}v = \rho^\varepsilon\theta^\varepsilon I_d, \qquad \int_{\mathbb{R}^d}|\omega|^2\,G\,\mathrm{d}v = d\rho^\varepsilon\theta^\varepsilon.$$

Expanding $c\otimes c$ and $|c|^2$ immediately gives (5.3).

For (5.4), we write

$$c\left(\frac{|c|^2}{\theta} - (d+2)\right) = (\omega + \delta u)\left(\frac{|\omega|^2 + 2\omega\cdot\delta u + |\delta u|^2}{\theta} - (d+2)\right).$$

After integrating against G we are left with

$$\begin{aligned}\int_{\mathbb{R}^d} c\left(\frac{|c|^2}{\theta} - (d+2)\right)G\,\mathrm{d}v &= \frac{\delta u}{\theta}\int_{\mathbb{R}^d}|\omega|^2 G\,\mathrm{d}v + \frac{2}{\theta}\int_{\mathbb{R}^d}\omega(\omega\cdot\delta u)G\,\mathrm{d}v \\ &\quad + \frac{|\delta u|^2}{\theta}\,\delta u\int_{\mathbb{R}^d} G\,\mathrm{d}v - (d+2)\delta u\int_{\mathbb{R}^d} G\,\mathrm{d}v.\end{aligned}$$

Using

$$\int_{\mathbb{R}^d}\omega(\omega\cdot\delta u)G\,\mathrm{d}v = \rho^\varepsilon\theta^\varepsilon\delta u, \qquad \int_{\mathbb{R}^d} G\,\mathrm{d}v = \rho^\varepsilon,$$

we obtain (5.4). □

Lemma 5.3. *Fix $\overline{r} > 0$. Then there exists $C = C(\overline{r}) > 0$ such that*

$$(r-1)^2 \le C\big(r - 1 - \log r\big) \qquad \textit{for all } r \in (0,\overline{r}].$$

Proof. The function $\psi(r) := r - 1 - \log r$ is C^2 and strictly convex on $(0,\infty)$, with $\psi(1) = \psi'(1) = 0$ and $\psi''(r) = 1/r$. By Taylor's theorem about the point $r = 1$, for every $r > 0$ there exists

$$\xi \in (\min\{1, r\}, \max\{1, r\})$$

such that

$$\psi(r) = \frac{1}{2}\,\psi''(\xi)(r-1)^2.$$

If $r \in (0, \overline{r}]$, then $\xi \le \max\{1, \overline{r}\}$, hence

$$\psi(r) \ge \frac{1}{2\max\{1,\overline{r}\}}(r-1)^2.$$

This proves the claim. □

6. Proof of Theorem 2.1

Throughout this proof, C denotes a positive constant, possibly changing from line to line, depending only on the quantities appearing in the statement of Theorem 2.1.

Set $c := v - u$, $G := \mathcal{M}(f^\varepsilon)$ and

$$D_\varepsilon(t) := \iint_{\mathbb{R}^{2d}} \big(f^\varepsilon - G\big) \log \frac{f^\varepsilon}{G}\,\mathrm{d}v\,\mathrm{d}x = H(f^\varepsilon \mid G) + H(G \mid f^\varepsilon) \ge 0. \tag{6.1}$$

By the relative entropy identity in Proposition 4.2,

$$\frac{\mathrm{d}}{\mathrm{d}t} H(f^\varepsilon \mid M) = -\frac{1}{\varepsilon} D_\varepsilon(t) + \mathcal{R}_u(t) + \mathcal{R}_\theta(t), \tag{6.2}$$

where

$$\mathcal{R}_u(t) := -\int_{\mathbb{R}^d} \frac{1}{\theta}\left(\int_{\mathbb{R}^d}\left(c\otimes c - \frac{|c|^2}{d} I_d\right) f^\varepsilon\,\mathrm{d}v\right) : \nabla_x u\,\mathrm{d}x,$$

$$\mathcal{R}_\theta(t) := -\frac{1}{2}\int_{\mathbb{R}^d} \nabla_x \log\theta \cdot \left(\int_{\mathbb{R}^d} c\left(\frac{|c|^2}{\theta} - (d+2)\right) f^\varepsilon\,\mathrm{d}v\right)\mathrm{d}x.$$

We split each remainder into a Maxwellian part and an extra term,

$$\mathcal{R}_u = \mathcal{R}_u^{\mathrm{M}} + \mathcal{R}_u^{\mathrm{E}}, \qquad \mathcal{R}_\theta = \mathcal{R}_\theta^{\mathrm{M}} + \mathcal{R}_\theta^{\mathrm{E}},$$

where the Maxwellian part is obtained by replacing f^ε with $G = \mathcal{M}(f^\varepsilon)$, and the extra term is obtained by replacing f^ε with $f^\varepsilon - G$.

Step 1: Maxwellian parts. Using Lemma 5.2, we obtain

$$\mathcal{R}_u^{\mathrm{M}} = -\int_{\mathbb{R}^d} \frac{\rho^\varepsilon}{\theta}\left(\delta u \otimes \delta u - \frac{|\delta u|^2}{d} I_d\right) : \nabla_x u\,\mathrm{d}x, \qquad \delta u := u^\varepsilon - u.$$

Since $\theta \ge \underline{\theta}$ and $\big|\delta u \otimes \delta u - (|\delta u|^2/d) I_d\big| \le 2|\delta u|^2$, we infer

$$|\mathcal{R}_u^{\mathrm{M}}| \le \frac{2}{\underline{\theta}} \|\nabla_x u\|_\infty \int_{\mathbb{R}^d} \rho^\varepsilon |\delta u|^2\,\mathrm{d}x. \tag{6.3}$$

On the other hand, Lemma 3.2 gives

$$\begin{aligned} H(G \mid M) = &\int_{\mathbb{R}^d} \rho^\varepsilon \log\frac{\rho^\varepsilon}{\rho}\,\mathrm{d}x + \frac{1}{2}\int_{\mathbb{R}^d} \rho^\varepsilon \frac{|\delta u|^2}{\theta}\,\mathrm{d}x \\ &+ \frac{d}{2}\int_{\mathbb{R}^d} \rho^\varepsilon\left(\frac{\theta^\varepsilon}{\theta} - 1 - \log\frac{\theta^\varepsilon}{\theta}\right)\mathrm{d}x, \end{aligned}$$

whence

$$\int_{\mathbb{R}^d} \rho^\varepsilon |\delta u|^2\,\mathrm{d}x \le 2\overline{\theta}\, H(G \mid M).$$

Therefore,

$$|\mathcal{R}_u^{\mathrm{M}}| \le C\, H(G \mid M), \tag{6.4}$$

where C depends only on $\underline{\theta}$, $\overline{\theta}$, and $\|\nabla_x u\|_\infty$.

Now, set $r := \theta^\varepsilon/\theta$. By Lemma 5.2,

$$\mathcal{R}_\theta^{\mathrm{M}} = -\frac{1}{2}\int_{\mathbb{R}^d} \rho^\varepsilon\, \delta u \cdot \nabla_x \log\theta \left((d+2)(r-1) + \frac{|\delta u|^2}{\theta} \right) \mathrm{d}x.$$

Hence

$$\begin{aligned} |\mathcal{R}_\theta^{\mathrm{M}}| &\le \frac{1}{2}\|\nabla_x \log\theta\|_\infty \int_{\mathbb{R}^d} \rho^\varepsilon |\delta u| \left((d+2)|r-1| + \frac{|\delta u|^2}{\theta} \right) \mathrm{d}x \\ &= \frac{1}{2}\|\nabla_x \log\theta\|_\infty \int_{\mathbb{R}^d} \rho^\varepsilon \left((d+2)|\delta u|\,|r-1| + \frac{|\delta u|^3}{\theta} \right) \mathrm{d}x. \end{aligned}$$

Moreover,

$$|\delta u| \le \|u^\varepsilon\|_\infty + \|u\|_\infty \le U + \|u\|_\infty.$$

Using $|ab| \le \frac{1}{2}(a^2+b^2)$ and $\theta \ge \underline{\theta}$, we obtain

$$|\delta u|\,|r-1| \le \frac{1}{2}\big(|\delta u|^2 + (r-1)^2\big),$$

and

$$\frac{|\delta u|^3}{\theta} \le \frac{1}{\underline{\theta}}|\delta u|^3 \le \frac{U + \|u\|_\infty}{\underline{\theta}}\,|\delta u|^2.$$

Substituting these bounds into the previous estimate yields

$$\begin{aligned} |\mathcal{R}_\theta^{\mathrm{M}}| &\le \frac{1}{2}\|\nabla_x \log\theta\|_\infty \int_{\mathbb{R}^d} \rho^\varepsilon \left(\frac{d+2}{2}\big(|\delta u|^2 + (r-1)^2\big) + \frac{U + \|u\|_\infty}{\underline{\theta}}|\delta u|^2 \right) \mathrm{d}x \\ &\le C\|\nabla_x \log\theta\|_\infty \int_{\mathbb{R}^d} \rho^\varepsilon \big(|\delta u|^2 + (r-1)^2\big)\mathrm{d}x, \end{aligned}$$

where C depends on d, $\underline{\theta}$, U, and $\|u\|_\infty$.

Since $0 < \theta^\varepsilon \le \overline{\Theta}$ and $\theta \ge \underline{\theta}$, the ratio $r = \theta^\varepsilon/\theta$ satisfies $0 < r \le \frac{\overline{\Theta}}{\underline{\theta}}$. Applying Lemma 5.3 with $\overline{r} = \overline{\Theta}/\underline{\theta}$, we obtain

$$(r-1)^2 \le C\,(r - 1 - \log r)\,,$$

where $C = C(\overline{\Theta}/\underline{\theta})$. Invoking again Lemma 3.2, we conclude that

$$|\mathcal{R}_\theta^{\mathrm{M}}| \le C\, H(G \mid M), \tag{6.5}$$

where now C depends on d, $\underline{\theta}$, $\overline{\theta}$, U, $\overline{\Theta}$, $\|u\|_\infty$, and $\|\nabla_x \log\theta\|_\infty$.

Combining (6.4) and (6.5) gives

$$|\mathcal{R}_u^{\mathrm{M}}| + |\mathcal{R}_\theta^{\mathrm{M}}| \le C\, H(G \mid M). \tag{6.6}$$

Step 2: Extra terms. Set $h := f^\varepsilon - G$. By definition of the extra terms, we have

$$\mathcal{R}_u^{\mathrm{E}} = -\iint_{\mathbb{R}^{2d}} \Phi_u\, h\, \mathrm{d}v\, \mathrm{d}x, \qquad \mathcal{R}_\theta^{\mathrm{E}} = -\iint_{\mathbb{R}^{2d}} \Phi_\theta\, h\, \mathrm{d}v\, \mathrm{d}x,$$

where

$$\Phi_u := \frac{1}{\theta}\left(c \otimes c - \frac{|c|^2}{d} I_d \right) : \nabla_x u, \qquad \Phi_\theta := \frac{1}{2}\,\nabla_x \log\theta \cdot c \left(\frac{|c|^2}{\theta} - (d+2) \right).$$

For any measurable Φ, we may write

$$\iint_{\mathbb{R}^{2d}} \Phi h\, \mathrm{d}v\, \mathrm{d}x = \iint_{\mathbb{R}^{2d}} \Big(\Phi\sqrt{f^\varepsilon + G}\Big)\Big(\frac{h}{\sqrt{f^\varepsilon + G}}\Big)\, \mathrm{d}v\, \mathrm{d}x,$$

and hence, by Cauchy–Schwarz,

$$\left|\iint_{\mathbb{R}^{2d}} \Phi h \,\mathrm{d}v\,\mathrm{d}x\right| \le \left(\iint_{\mathbb{R}^{2d}} \Phi^2 (f^\varepsilon + G)\,\mathrm{d}v\,\mathrm{d}x\right)^{1/2} \left(\iint_{\mathbb{R}^{2d}} \frac{h^2}{f^\varepsilon + G}\,\mathrm{d}v\,\mathrm{d}x\right)^{1/2}, \quad (6.7)$$

with the convention $h^2/(f^\varepsilon + G) = 0$ on $\{f^\varepsilon + G = 0\}$.

By Lemma 5.1 and (6.1),

$$\iint_{\mathbb{R}^{2d}} \frac{h^2}{f^\varepsilon + G}\,\mathrm{d}v\,\mathrm{d}x \le \frac{1}{2} D_\varepsilon(t). \quad (6.8)$$

We next estimate the growth of Φ_u and Φ_θ in v. Since

$$|c| = |v - u| \le |v| + \|u\|_\infty,$$

and since $0 < \underline{\theta} \le \theta \le \overline{\theta}$, we have

$$\begin{aligned} |\Phi_u| &\le \frac{\|\nabla_x u\|_\infty}{\underline{\theta}} \left| c \otimes c - \frac{|c|^2}{d} I_d \right| \\ &\le C|c|^2 \\ &\le C(1 + |v|^2). \end{aligned}$$

Similarly,

$$\begin{aligned} |\Phi_\theta| &\le \frac{1}{2} \|\nabla_x \log\theta\|_\infty\, |c| \left(\frac{|c|^2}{\underline{\theta}} + d + 2 \right) \\ &\le C\big(|c| + |c|^3\big) \\ &\le C(1 + |v|^3), \end{aligned}$$

where in the last step we used the inequality $|v| \le 1 + |v|^3$.

Therefore,

$$\Phi_u^2 + \Phi_\theta^2 \le C\big(1 + |v|^6\big), \quad (6.9)$$

where C depends only on d, $\underline{\theta}$, $\|u\|_\infty$, $\|\nabla_x u\|_\infty$, and $\|\nabla_x \log\theta\|_\infty$.

Using (6.9), the unit mass of f^ε, and the assumed sixth moment bound, we get

$$\iint_{\mathbb{R}^{2d}} (\Phi_u^2 + \Phi_\theta^2) f^\varepsilon \,\mathrm{d}v\,\mathrm{d}x \le C \iint_{\mathbb{R}^{2d}} (1 + |v|^6) f^\varepsilon \,\mathrm{d}v\,\mathrm{d}x \le C.$$

We now estimate the same quantity with G. Writing $\omega := v - u^\varepsilon$, we have

$$G = \frac{\rho^\varepsilon}{(2\pi\theta^\varepsilon)^{d/2}} \exp\left(-\frac{|\omega|^2}{2\theta^\varepsilon}\right).$$

Hence

$$\begin{aligned} \int_{\mathbb{R}^d} |v|^6 G\,\mathrm{d}v &= \frac{\rho^\varepsilon}{(2\pi\theta^\varepsilon)^{d/2}} \int_{\mathbb{R}^d} |\omega + u^\varepsilon|^6 \exp\left(-\frac{|\omega|^2}{2\theta^\varepsilon}\right) \mathrm{d}\omega \\ &\le C\, \frac{\rho^\varepsilon}{(2\pi\theta^\varepsilon)^{d/2}} \int_{\mathbb{R}^d} \big(|\omega|^6 + |u^\varepsilon|^6\big) \exp\left(-\frac{|\omega|^2}{2\theta^\varepsilon}\right) \mathrm{d}\omega, \end{aligned}$$

where we used the elementary inequality $|a + b|^6 \le C\big(|a|^6 + |b|^6\big)$. Since $|u^\varepsilon| \le U$ and $\theta^\varepsilon \le \overline{\Theta}$, a change of variables $\omega = \sqrt{\theta^\varepsilon} z$ shows that

$$\frac{1}{(2\pi\theta^\varepsilon)^{d/2}} \int_{\mathbb{R}^d} |\omega|^6 \exp\left(-\frac{|\omega|^2}{2\theta^\varepsilon}\right) \mathrm{d}\omega = (\theta^\varepsilon)^3 (2\pi)^{-d/2} \int_{\mathbb{R}^d} |z|^6 e^{-|z|^2/2}\,\mathrm{d}z \le C,$$

and also

$$\frac{1}{(2\pi\theta^\varepsilon)^{d/2}} \int_{\mathbb{R}^d} \exp\left(-\frac{|\omega|^2}{2\theta^\varepsilon}\right) \mathrm{d}\omega = 1.$$

Therefore

$$\int_{\mathbb{R}^d} |v|^6 G \,\mathrm{d}v \le C\,\rho^\varepsilon,$$

with $C = C(d, U, \overline{\Theta})$. Using (6.9) again and $\int_{\mathbb{R}^d} \rho^\varepsilon \,\mathrm{d}x = 1$, we conclude that

$$\iint_{\mathbb{R}^{2d}} (\Phi_u^2 + \Phi_\theta^2) G \,\mathrm{d}v \,\mathrm{d}x \le C.$$

Combining the previous two bounds, we obtain

$$\iint_{\mathbb{R}^{2d}} (\Phi_u^2 + \Phi_\theta^2)(f^\varepsilon + G) \,\mathrm{d}v \,\mathrm{d}x \le C,$$

for some constant C independent of ε and t. Applying (6.7) with $\Phi = \Phi_u$ and $\Phi = \Phi_\theta$, and using (6.8), we infer that

$$\begin{aligned} |\mathcal{R}_u^{\mathrm{E}}| + |\mathcal{R}_\theta^{\mathrm{E}}| &\le \left(\iint_{\mathbb{R}^{2d}} \Phi_u^2 (f^\varepsilon + G) \,\mathrm{d}v \,\mathrm{d}x \right)^{1/2} \left(\frac{1}{2} D_\varepsilon(t) \right)^{1/2} \\ &\quad + \left(\iint_{\mathbb{R}^{2d}} \Phi_\theta^2 (f^\varepsilon + G) \,\mathrm{d}v \,\mathrm{d}x \right)^{1/2} \left(\frac{1}{2} D_\varepsilon(t) \right)^{1/2} \\ &\le C \sqrt{D_\varepsilon(t)}. \end{aligned}$$

Finally, Young's inequality gives

$$|\mathcal{R}_u^{\mathrm{E}}| + |\mathcal{R}_\theta^{\mathrm{E}}| \le \frac{1}{2\varepsilon} D_\varepsilon(t) + \frac{\varepsilon}{2} C. \tag{6.10}$$

Step 3: Gronwall estimate. Combining (6.2), (6.6), and (6.10), we obtain

$$\frac{\mathrm{d}}{\mathrm{d}t} H(f^\varepsilon \mid M) \le -\frac{1}{\varepsilon} D_\varepsilon(t) + \frac{1}{2\varepsilon} D_\varepsilon(t) + C\, H(G \mid M) + C\varepsilon.$$

By Lemma 3.1,

$$H(f^\varepsilon \mid M) = H(f^\varepsilon \mid G) + H(G \mid M),$$

so that $H(G \mid M) \le H(f^\varepsilon \mid M)$. Therefore,

$$\frac{\mathrm{d}}{\mathrm{d}t} H(f^\varepsilon \mid M) \le C\, H(f^\varepsilon \mid M) + C\varepsilon, \tag{6.11}$$

establishing (2.1). Gronwall's lemma then yields

$$\sup_{t \in [0,T)} H\big(f^\varepsilon(t) \mid M(t)\big) \le e^{CT} \big(H(f_0^\varepsilon \mid M_0) + \varepsilon \big). \tag{6.12}$$

By the well-preparedness assumption (2.2) it follows from (6.12) that

$$\sup_{t \in [0,T)} H\big(f^\varepsilon(t) \mid M(t)\big) \to 0 \qquad \text{as } \varepsilon \to 0,$$

which finishes the proof. □

Remark 6.1. Since M_0 is a Maxwellian, Lemma 3.1 gives

$$H(f_0^\varepsilon \mid M_0) = H\big(f_0^\varepsilon \mid \mathcal{M}(f_0^\varepsilon)\big) + H\big(\mathcal{M}(f_0^\varepsilon) \mid M_0\big).$$

In particular, the well-preparedness condition

$$H(f_0^\varepsilon \mid M_0) \to 0 \qquad \text{as } \varepsilon \to 0$$

is equivalent to

$$H\big(f_0^\varepsilon \mid \mathcal{M}(f_0^\varepsilon)\big) \to 0, \qquad H\big(\mathcal{M}(f_0^\varepsilon) \mid M_0\big) \to 0, \qquad \text{as } \varepsilon \to 0.$$

7. Proof of Corollary 2.2

We first prove the convergence in phase space. Since both $f^\varepsilon(t)$ and $M(t)$ have unit mass for every $t \in [0,T)$, the Csiszár–Kullback–Pinsker inequality gives

$$\|f^\varepsilon(t) - M(t)\|^2_{L^1(\mathbb{R}^{2d})} \le 2\, H\big(f^\varepsilon(t) \mid M(t)\big).$$

Taking the supremum over $t \in [0,T)$ and using (2.3), we obtain

$$f^\varepsilon \to M \qquad \text{in } L^\infty(0,T;L^1(\mathbb{R}^{2d})).$$

Moreover, since $M(t)$ is a Maxwellian for every $t \in [0,T)$, Lemma 3.1 yields

$$H\big(f^\varepsilon(t) \mid M(t)\big) = H\big(f^\varepsilon(t) \mid \mathcal{M}(f^\varepsilon)(t)\big) + H\big(\mathcal{M}(f^\varepsilon)(t) \mid M(t)\big).$$

Hence

$$H\big(\mathcal{M}(f^\varepsilon)(t) \mid M(t)\big) \le H\big(f^\varepsilon(t) \mid M(t)\big) \qquad \text{for all } t \in [0,T).$$

Since $\mathcal{M}(f^\varepsilon)(t)$ and $M(t)$ also have unit mass, the Csiszár–Kullback–Pinsker inequality gives

$$\|\mathcal{M}(f^\varepsilon)(t) - M(t)\|^2_{L^1(\mathbb{R}^{2d})} \le 2\, H\big(\mathcal{M}(f^\varepsilon)(t) \mid M(t)\big) \le 2\, H\big(f^\varepsilon(t) \mid M(t)\big).$$

Taking the supremum in time and using (2.3), we conclude that

$$\mathcal{M}(f^\varepsilon) \to M \qquad \text{in } L^\infty(0,T;L^1(\mathbb{R}^{2d})).$$

We now turn to the macroscopic quantities. Since both $\mathcal{M}(f^\varepsilon)$ and M are Maxwellians of the form (1.8), Lemma 3.2 gives

$$\begin{aligned} H(\mathcal{M}(f^\varepsilon) \mid M) = &\int_{\mathbb{R}^d} \rho^\varepsilon \log \frac{\rho^\varepsilon}{\rho}\, \mathrm{d}x + \frac{1}{2}\int_{\mathbb{R}^d} \rho^\varepsilon \frac{|u^\varepsilon - u|^2}{\theta}\, \mathrm{d}x \\ &+ \frac{d}{2}\int_{\mathbb{R}^d} \rho^\varepsilon \left(\frac{\theta^\varepsilon}{\theta} - 1 - \log\frac{\theta^\varepsilon}{\theta}\right) \mathrm{d}x. \end{aligned} \tag{7.1}$$

Since all terms on the right-hand side of (7.1) are nonnegative, while

$$H\big(\mathcal{M}(f^\varepsilon)(t) \mid M(t)\big) \le H\big(f^\varepsilon(t) \mid M(t)\big) \qquad \text{for all } t \in [0,T),$$

it follows from (2.3) that

$$\sup_{t\in[0,T)} \int_{\mathbb{R}^d} \rho^\varepsilon(t) \log \frac{\rho^\varepsilon(t)}{\rho(t)}\, \mathrm{d}x \to 0, \tag{7.2}$$

$$\sup_{t\in[0,T)} \int_{\mathbb{R}^d} \rho^\varepsilon(t) \frac{|u^\varepsilon(t) - u(t)|^2}{\theta(t)}\, \mathrm{d}x \to 0, \tag{7.3}$$

and

$$\sup_{t\in[0,T)} \int_{\mathbb{R}^d} \rho^\varepsilon(t) \left(\frac{\theta^\varepsilon(t)}{\theta(t)} - 1 - \log\frac{\theta^\varepsilon(t)}{\theta(t)}\right) \mathrm{d}x \to 0. \tag{7.4}$$

Applying the Csiszár–Kullback–Pinsker inequality on $\mathbb{R}^d$ to the probability densities $\rho^\varepsilon(t)$ and $\rho(t)$, we deduce from (7.2) that

$$\rho^\varepsilon \to \rho \qquad \text{in } L^\infty(0,T;L^1(\mathbb{R}^d)).$$

Next,

$$\begin{aligned}\|\rho^\varepsilon u^\varepsilon - \rho u\|_{L^1(\mathbb{R}^d)} &\le \|\rho^\varepsilon(u^\varepsilon - u)\|_{L^1(\mathbb{R}^d)} + \|u\|_\infty \|\rho^\varepsilon - \rho\|_{L^1(\mathbb{R}^d)} \\ &\le \left(\int_{\mathbb{R}^d} \rho^\varepsilon \,\mathrm{d}x\right)^{1/2} \left(\int_{\mathbb{R}^d} \rho^\varepsilon |u^\varepsilon - u|^2 \,\mathrm{d}x\right)^{1/2} \\ &\quad + \|u\|_\infty \|\rho^\varepsilon - \rho\|_{L^1(\mathbb{R}^d)} \\ &\le \overline{\theta}^{1/2} \left(\int_{\mathbb{R}^d} \rho^\varepsilon \frac{|u^\varepsilon - u|^2}{\theta} \,\mathrm{d}x\right)^{1/2} \\ &\quad + \|u\|_\infty \|\rho^\varepsilon - \rho\|_{L^1(\mathbb{R}^d)}.\end{aligned}$$

Since $\int_{\mathbb{R}^d} \rho^\varepsilon \,\mathrm{d}x = 1$ and both terms on the right-hand side converge to zero uniformly in time by (7.3) and the convergence of ρ^ε, we obtain

$$\rho^\varepsilon u^\varepsilon \to \rho u \qquad \text{in } L^\infty(0,T;L^1(\mathbb{R}^d)).$$

Finally, since $0 < \theta^\varepsilon \le \overline{\Theta}$ and $\theta \ge \underline{\theta}$, we have

$$0 < \frac{\theta^\varepsilon}{\theta} \le \frac{\overline{\Theta}}{\underline{\theta}}.$$

Applying Lemma 5.3 with

$$r = \frac{\theta^\varepsilon}{\theta},$$

we infer that there exists a constant $C = C(\overline{\Theta}/\underline{\theta}) > 0$ such that

$$\left(\frac{\theta^\varepsilon}{\theta} - 1\right)^2 \le C\left(\frac{\theta^\varepsilon}{\theta} - 1 - \log\frac{\theta^\varepsilon}{\theta}\right).$$

Since $\theta \le \overline{\theta}$, multiplying by $\rho^\varepsilon \theta^2$ and integrating yields

$$\int_{\mathbb{R}^d} \rho^\varepsilon |\theta^\varepsilon - \theta|^2 \,\mathrm{d}x \le C\,\overline{\theta}^2 \int_{\mathbb{R}^d} \rho^\varepsilon \left(\frac{\theta^\varepsilon}{\theta} - 1 - \log\frac{\theta^\varepsilon}{\theta}\right) \mathrm{d}x.$$

Hence (7.4) implies

$$\sup_{t\in[0,T)} \int_{\mathbb{R}^d} \rho^\varepsilon(t) |\theta^\varepsilon(t) - \theta(t)|^2 \,\mathrm{d}x \to 0.$$

Therefore,

$$\begin{aligned}\|\rho^\varepsilon \theta^\varepsilon - \rho\theta\|_{L^1(\mathbb{R}^d)} &\le \|\rho^\varepsilon(\theta^\varepsilon - \theta)\|_{L^1(\mathbb{R}^d)} + \|\theta(\rho^\varepsilon - \rho)\|_{L^1(\mathbb{R}^d)} \\ &\le \left(\int_{\mathbb{R}^d} \rho^\varepsilon \,\mathrm{d}x\right)^{1/2} \left(\int_{\mathbb{R}^d} \rho^\varepsilon |\theta^\varepsilon - \theta|^2 \,\mathrm{d}x\right)^{1/2} \\ &\quad + \|\theta\|_\infty \|\rho^\varepsilon - \rho\|_{L^1(\mathbb{R}^d)}.\end{aligned}$$

Since $\int_{\mathbb{R}^d} \rho^\varepsilon \,\mathrm{d}x = 1$, both terms on the right-hand side converge to zero uniformly in time, so we conclude that

$$\rho^\varepsilon \theta^\varepsilon \to \rho\theta \qquad \text{in } L^\infty(0,T;L^1(\mathbb{R}^d)).$$

This completes the proof. □

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References

[1] F. Berthelin and F. Bouchut, Solution with finite energy to a BGK system relaxing to isentropic gas dynamics, *Ann. Fac. Sci. Toulouse Math.* **9** (2000), no. 4, 605–630.

[2] F. Berthelin and F. Bouchut, Relaxation to isentropic gas dynamics for a BGK system with single kinetic entropy, *Methods Appl. Anal.* **9** (2002), 313–328.

[3] F. Berthelin and F. Bouchut, Kinetic invariant domains and relaxation limit from a BGK model to isentropic gas dynamics, *Asymptot. Anal.* **31** (2002), 153–176.

[4] F. Berthelin and A. Vasseur, From kinetic equations to multidimensional isentropic gas dynamics before shocks, *SIAM J. Math. Anal.* **36** (2005), 1807–1835.

[5] F. Berthelin, A. E. Tzavaras and A. Vasseur, From discrete velocity Boltzmann equations to gas dynamics before shocks, *J. Stat. Phys.* **135** (2009), 153–173.

[6] P. L. Bhatnagar, E. P. Gross and M. Krook, A model for collision processes in gases. I. Small amplitude processes in charged and neutral one-component systems, *Phys. Rev.* **94** (1954), 511–525.

[7] Y.-P. Choi and B.-H. Hwang, Global existence of weak solutions to a BGK model relaxing to the barotropic Euler equations, *Nonlinear Anal.* **238** (2024), 113414.

[8] Y.-P. Choi and B.-H. Hwang, From BGK-alignment model to the pressured Euler-alignment system with singular communication weights, *J. Differential Equations* **379** (2024), 363–412.

[9] C. Dafermos, The second law of thermodynamics and stability, *Arch. Ration. Mech. Anal.* **70** (1979), 167–179.

[10] G. L. Gilardoni, On Pinsker's and Vajda's type inequalities for Csiszar's f-divergences, *IEEE Trans. Inform. Theory* **56** (2010), 5377–5386.

[11] B.-H. Hwang, Classical solutions to a BGK-type model relaxing to the isentropic gas dynamics, *Bull. Korean Math. Soc.* **62** (2025), no. 1, 217–236.

[12] D. Koo and S. Song, Global mild solutions to a BGK model for barotropic gas dynamics, *SIAM J. Math. Anal.* **57** (2025), no. 4, 4137–4164.

[13] B. Perthame, Global existence to the BGK model of Boltzmann equation, *J. Differential Equations* **82** (1989), no. 1, 191–205.

[14] A. E. Tzavaras, On the mathematical theory of fluid dynamic limits to conservation laws, in *Advances in Mathematical Fluid Mechanics (Paseky, 1999)*, Springer, Berlin, 2000, pp. 192–222.

[15] A. E. Tzavaras, Relative entropy in hyperbolic relaxation, *Commun. Math. Sci.* **3** (2005), no. 2, 119–132.

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